

PRANAV SINGH

Full Stack Developer

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TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, C++,

Frontend: HTML, CSS, React, Next.js, Tailwind CSS

Backend & Databases: Node.js, Flask, REST APIs, MySQL, Firebase (Firestore, Realtime Database, Auth)

AI & Tooling: LLM/AI API integration (Gemini, Claude), prompt engineering, Git/GitHub, CI/CD (GitHub Actions)

PROJECTS

Sprinto — Real-Time Collaborative Workspace & Kanban Platform

Next.js, TypeScript, Firebase (Firestore, Auth), Tailwind CSS, @dnd-kit, IndexedDB

Live Demo: <https://sprinto-move.vercel.app/> | **GitHub:** <https://github.com/PRANAV-SINGH-CSE/Sprinto>

- Built a real-time collaborative Kanban platform that lets multiple users edit the same board simultaneously without conflicts, replacing manual refreshes with live Firestore-backed sync.
- Profiled the full sync pipeline: optimistic UI updates render in under 1ms locally, while changes propagate to other clients in ~4.5ms median over Firestore's WebSocket channel, decoupled from the ~350ms average cloud write commit.
- Cut repeat-visit board load time from ~400ms (Firestore query) to ~5ms using an IndexedDB local cache layer, keeping the board responsive even on unstable connections.

LiveVoice AI — Real-Time Speech Intelligence & Desktop HUD Platform

Python, FastAPI, WebSockets, AsyncIO, PyQt5, Win32 API, JavaScript (ES6+), HTML5, CSS3, OpenAI API, Anthropic API, Faster-Whisper

- Built a real-time voice assistant that transcribes speech and streams AI responses live, achieving sub-700ms latency from end of speech to first spoken token and a sub-1-second full response loop, measured via P50/P90 profiling across the audio-to-LLM pipeline.
- Tuned the speech-to-text stage to run at a 0.28x real-time factor (3.5x faster than speech) with Faster-Whisper, while the LLM streamed responses at ~197 tokens/sec.
- Designed a transparent, always-on-top desktop overlay controlled by global hotkeys, usable during screen shares and calls without disrupting other applications.
- Built an async FastAPI backend that extracts key questions from streamed transcripts in real time, with full session history exportable to Markdown/JSON.

ZentiqAI — AI Chat Web Application

Python (Flask), JavaScript, Firebase (Firestore, Realtime Database), HTML, CSS

Live Demo: <https://zentiqai.vercel.app/> | **GitHub:** https://github.com/PRANAV-SINGH-CSE/ZentiqAI_Chat_Bot

- Built a full stack AI chat app with persistent, multi-session conversations and image upload, letting users pick up a conversation from any device.
- Engineered concurrent session handling on the backend so multiple users can chat at the same time without their session data colliding.
- Rebuilt the frontend into a modular, class-based JavaScript architecture, making it straightforward to add new chat features without touching unrelated code.

EDUCATION

Lovely Professional University (LPU)

B.Tech, Computer Science and Engineering